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Antisymmetric Spin Interaction in Metals

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The bilinear term of the spin-polarization density in the spin-dependent energy of metals is separated into the terms of the symmetric spin interaction (SSI) and of the antisymmetric spin interaction (ASI), whose properties are first investigated phenomenologically. A microscopic derivation of the SSI and ASI is given for a metal which has no inversion center. It is shown that the spin-orbit coupling which splits electronic bands in the absence of inversion center produces an antisymmetric component of the wave-vector-dependent susceptibility and the ASI is proportional to the antisymmetric component to the first order of the latter quantity. The obtained ASI energy is concluded to be not always small compared with the SSI energy.

§1. Introduction

Recently, it was found that the antisymmetric spin interaction (ASI) operating in some metallic compounds of low crystal symmetry plays an important role in determining the spin structure. MnSi, for example, has a metallic conductivity and a weak magnetic moment 0.4 μ_B/Mn which is much smaller than that expected for ionic states of Mn.[1-3] This compound, therefore, is considered to be a typical example of the itinerant electron magnets. Ishikawa *et al.*[4] found by neutron diffraction experiment that MnSi has a helical spin structure with a very long period independent of the direction of its propagation vector. This and other peculiar properties have been successfully explained by the assumption that the ferromagnetic state becomes unstable toward the helical spin structure because of the presence of the ASI.[5-7] The same mechanism operates also in FeGe,[8] $Fe_xCo_{1-x}Si$, and $Co_xMn_{1-x}Si$.[9,10] Another example of such compounds with the ASI is Mn_3Sn, which also has a metallic conductivity[11] and a small magnetic moment 1.78 μ_B/Mn.[12] In this compound, the inverse triangular spin structure is realized by the presence of the ASI, although the symmetric spin interaction (SSI) (including the magnetic anisotropy energy) favors the normal triangular structure.[13] Isomorphous Mn_3Ge exhibits similar properties to those of Mn_3Sn.[14]

Dzyaloshinsky[15] first recognized the existence of the antisymmetric spin interaction in magnetic insulators. He wrote down the free energy of a spin system in terms of spins well localized on magnetic ions and required every term in the expression of the free energy to satisfy the full crystal symmetry. As the result, the antisymmetric form was shown to be allowed in a certain class of crystals with sufficiently low symmetries. The derivation of the ASI on the basis of a microscopic Hamiltonian was given by Moriya.[16] Moriya extends the theory of the superexchange in ionic crystals to include the spin-orbit coupling, assuming that the transfer integrals of electrons between magnetic ions are sufficiently small compared with the intraatomic Coulomb energy and the superexchange (or direct exchange) interaction between electrons on neighboring magnetic ions are sufficiently small compared with the crystal field splitting. The interplay of the superexchange (or direct exchange) interaction and the spin-orbit coupling was shown to cause the ASI. The two above theories, however, can not be applied to itinerant electron systems, since the assumptions employed do not hold valid in metals. The purpose of this paper is to show how the ASI in metals is derived from a microscopic point of view.*

The spin-dependent energy of a metal is a functional of a spin density varying with the position. We consider bilinear terms which appear in the expansion of the spin-dependent energy in powers of the spin density. In general, the bilinear terms can be classified into two parts, symmetric one and antisymmetric one with respect to the interchange of two spin densities. The fact that the antisymmetric part reflects the crystallographic symmetry suggests a role of the spin-orbit coupling in generating spin polarizations in band electron systems. Especially in a metal with no inversion center, the spin-orbit coupling lifts the spin degeneracy of electronic bands, giving rise to the energy of the ASI. In this paper, we confine ourselves to the derivation of the ASI in such a metal.

* A preliminary report presented at the International Conference on Magnetism 1982 (O. Nakanishi, M. Kataoka and A. Yanase, J. Magn. & Magn. Mater. **31–34** (1983) 339) treated the same problem but the ASI obtained there was found to be incorrect. The present paper gives also its revision.

In the next section, we investigate the properties of the SSI and the ASI in a general metal by use of a phenomenological argument. The microscopic Hamiltonian for a metal with no inversion center is obtained in §3. In §4, we calculate the wave-vector-dependent susceptibility of the metal. This quantity plays an important role in the calculation of the bilinear terms of the spin density in the spin-dependent energy as will be seen in §5. In §5, the bilinear terms are calculated to be separated into the terms of the SSI and of the ASI. An order of magnitude estimate of the ASI in MnSi is given there also. Concluding remarks are given in §6.

§2. Properties of the Symmetric Spin Interaction (SSI) and the Antisymmetric Spin Interaction (ASI)

Before going into details of the microscopic calculation of the spin interaction, we here investigate the properties of the SSI and the ASI. A metal under consideration is assumed to have a static spin polarization with a spatially varying magnitude. The spin density in the unit of $\hbar$ at the position $\boldsymbol{r}=\boldsymbol{R}_n+\boldsymbol{\rho}$ is denoted by $\boldsymbol{S}(\boldsymbol{R}_n+\boldsymbol{\rho})$ where $\boldsymbol{R}_n$ represents the position of the n-th unit cell and $\boldsymbol{\rho}$ represents a position in a unit cell. When the spin-dependent energy is expanded in powers of the spin density, the bilinear term is written as

$$\mathscr{E}=\sum_{nn'}\sum_{ii'}\int_{\mathrm{c}} S_i(\boldsymbol{R}_n+\boldsymbol{\rho})K_{ii'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho}\boldsymbol{\rho}')S_{i'}(\boldsymbol{R}_{n'}+\boldsymbol{\rho}')\,\mathrm{d}\boldsymbol{\rho}\,\mathrm{d}\boldsymbol{\rho}',\tag{1}$$

where i and i' specify the x, y and z components of quantities, $K_{ii'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho}\boldsymbol{\rho}')$ is the interaction kernel between $S_i(\boldsymbol{R}_n+\boldsymbol{\rho})$ and $S_{i'}(\boldsymbol{R}_{n'}+\boldsymbol{\rho}')$, and the subscript c means the integrations within unit cells. It should be noted that $K_{ii'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho}\boldsymbol{\rho}')$ is nonlocal in general but depends on $(\boldsymbol{R}_n-\boldsymbol{R}_{n'})$ because of the lattice periodicity. The interaction kernel has the symmetry given by

$$K_{ii'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho}\boldsymbol{\rho}')=K_{i'i}(\boldsymbol{R}_{n'}-\boldsymbol{R}_n;\,\boldsymbol{\rho}'\boldsymbol{\rho}).\tag{2}$$

The crystal symmetries require some additional relations between the kernels. If, for example, the metal has an inversion center, we have

$$K_{ii'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho}\boldsymbol{\rho}')=K_{ii'}(\boldsymbol{R}_{n'}-\boldsymbol{R}_n;\,-\boldsymbol{\rho}-\boldsymbol{\rho}'),\tag{3}$$

where an inversion center has been chosen for the origin of the coordinate. If the metal has the reflection plane which bisects the angle between the i and i' planes, we have

$$K_{ii'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho}\boldsymbol{\rho}')=K_{i'i}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho}\boldsymbol{\rho}').\tag{4}$$

We introduce $J_{ii'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho}\boldsymbol{\rho}')$ and $D_i(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho}\boldsymbol{\rho}')$ which are defined by

$$J_{ii'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')=\frac{1}{2}[K_{ii'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')+K_{i'i}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')], \tag{5}$$

$$D_i(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')=\frac{1}{2}[K_{i'i''}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')-K_{i''i'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')], \tag{6}$$

where a set of i, i' and i'' is obtained by a cyclic permutation of x, y and z. The quantities $J_{ii'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')$ and $D_{i''}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')$ are symmetric and antisymmetric with respect to the interchange of i and i'. Then, eq. (1) is rewritten as

$$\begin{aligned}\mathscr{E}=&\sum_{nn'}\sum_{ii'}\int_{\mathrm{c}} S_i(\boldsymbol{R}_n+\boldsymbol{\rho})J_{ii'}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')S_{i'}(\boldsymbol{R}_{n'}+\boldsymbol{\rho}')\,\mathrm{d}\boldsymbol{\rho}\,\mathrm{d}\boldsymbol{\rho}'\\ &+\sum_{nn'}\sum_{i}\int_{\mathrm{c}} D_i(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')[\boldsymbol{S}(\boldsymbol{R}_n+\boldsymbol{\rho})\times\boldsymbol{S}(\boldsymbol{R}_{n'}+\boldsymbol{\rho}')]_i\,\mathrm{d}\boldsymbol{\rho}\,\mathrm{d}\boldsymbol{\rho}'.\end{aligned} \tag{7}$$

The first term on the right hand side of eq. (7) represents the SSI including the usual exchange interaction, while the second term represents the ASI. From eqs. (2)–(4) and (6), we see that only $\boldsymbol{D}(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')$ with $\boldsymbol{\rho}=-\boldsymbol{\rho}'$ vanishes if the metal has an inversion center and that $D_i(\boldsymbol{R}_n-\boldsymbol{R}_{n'};\,\boldsymbol{\rho\rho}')$ also vanishes if the metal has the reflection plane which bisects the angle between the i' and i'' planes.

The spin density $S_i(\boldsymbol{R}_n+\boldsymbol{\rho})$ is transformed into its Fourier components $S_{\boldsymbol{G},i}(\boldsymbol{q})$:

$$S_i(\boldsymbol{R}_n+\boldsymbol{\rho})=\sum_{\boldsymbol{q},\boldsymbol{G}} S_{\boldsymbol{G},i}(\boldsymbol{q})\exp\{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_n+\mathrm{i}(\boldsymbol{q}+\boldsymbol{G})\cdot\boldsymbol{\rho}\}, \tag{8}$$

where $\boldsymbol{G}$ is a reciprocal lattice vector and $\boldsymbol{q}$ is a vector in the first Brillouin zone. Substituting eq. (8) into eq. (7), we obtain

$$\mathscr{E}=\Omega\sum_{\boldsymbol{GG}'}\sum_{\boldsymbol{q}}\{\sum_{ii'}S_{\boldsymbol{G},i}(\boldsymbol{q})J_{\boldsymbol{GG}',ii'}(\boldsymbol{q})S_{\boldsymbol{G}',i'}(\boldsymbol{q})^*+\sum_{i}D_{\boldsymbol{GG}',i}(\boldsymbol{q})[\boldsymbol{S}_{\boldsymbol{G}}(\boldsymbol{q})\times\boldsymbol{S}_{\boldsymbol{G}'}(\boldsymbol{q})^*]_i\}, \tag{9}$$

with

$$J_{\boldsymbol{GG}',ii'}(\boldsymbol{q})=\frac{1}{\Omega_{\mathrm{c}}}\sum_{n}\int_{\mathrm{c}} J_{ii'}(\boldsymbol{R}_n;\,\boldsymbol{\rho\rho}')\exp\{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_n+\mathrm{i}(\boldsymbol{q}+\boldsymbol{G})\cdot\boldsymbol{\rho}-\mathrm{i}(\boldsymbol{q}+\boldsymbol{G}')\cdot\boldsymbol{\rho}'\}\,\mathrm{d}\boldsymbol{\rho}\,\mathrm{d}\boldsymbol{\rho}', \tag{10}$$

$$D_{\boldsymbol{GG}',i}(\boldsymbol{q})=\frac{1}{\Omega_{\mathrm{c}}}\sum_{n}\int_{\mathrm{c}} D_i(\boldsymbol{R}_n;\,\boldsymbol{\rho\rho}')\exp\{\mathrm{i}\boldsymbol{q}\cdot\boldsymbol{R}_n+\mathrm{i}(\boldsymbol{q}+\boldsymbol{G})\cdot\boldsymbol{\rho}-\mathrm{i}(\boldsymbol{q}+\boldsymbol{G}')\cdot\boldsymbol{\rho}'\}\,\mathrm{d}\boldsymbol{\rho}\,\mathrm{d}\boldsymbol{\rho}', \tag{11}$$

where Ω and Ω_{c} are respectively the volumes of the crystal and of a unit cell. Equations (5), (6), (10) and (11) give the relations as

$$J_{\boldsymbol{GG}',ii'}(\boldsymbol{q})(=J_{\boldsymbol{GG}',i'i}(\boldsymbol{q}))=J_{\boldsymbol{G}'\boldsymbol{G},i'i}(\boldsymbol{q})^*=J_{-\boldsymbol{G}'-\boldsymbol{G},ii'}(-\boldsymbol{q}), \tag{12}$$

$$D_{\boldsymbol{GG}',i}(\boldsymbol{q})=-D_{\boldsymbol{G}'\boldsymbol{G},i}(\boldsymbol{q})^*=-D_{-\boldsymbol{G}'-\boldsymbol{G},i}(-\boldsymbol{q}). \tag{13}$$

If the metal has an inversion center, both $J_{\boldsymbol{GG}',ii'}(\boldsymbol{q})$ and $D_{\boldsymbol{GG}',i}(\boldsymbol{q})$ can be proved to be real quantities. Other symmetry operations belonging to the group of wave vector give additional relations between the quantities $J_{\boldsymbol{GG}',ii'}(\boldsymbol{q})$ or $D_{\boldsymbol{GG}',i}(\boldsymbol{q})$. In this paper, we do not go into detail of this problem.

We are often interested in a spin structure described by only one wave vector $\boldsymbol{q}$ with a small magnitude in a metal with no inversion center. Such spin structures are found in some compounds with the B20 type of crystal structure including MnSi. Then, the spin density $\boldsymbol{S}(\boldsymbol{r})$ is expressed by

$$\boldsymbol{S}(\boldsymbol{r})=\boldsymbol{S}_1\cos\boldsymbol{q}\cdot\boldsymbol{r}+\boldsymbol{S}_2\sin\boldsymbol{q}\cdot\boldsymbol{r}, \tag{14}$$

where

$$\begin{aligned}\boldsymbol{S}_1&=\boldsymbol{S}_0(\boldsymbol{q})+\boldsymbol{S}_0(-\boldsymbol{q}),\\ \boldsymbol{S}_2&=\mathrm{i}[\boldsymbol{S}_0(\boldsymbol{q})-\boldsymbol{S}_0(-\boldsymbol{q})].\end{aligned} \tag{15}$$

On the other hand, eqs. (12) and (13) give

$$J_{00,ii'}(\boldsymbol{q})\simeq J_{00,ii'}(0)\equiv J_{ii'}, \tag{16}$$

$$D_{00,i}(\boldsymbol{q})\simeq -\mathrm{i}\sum_{i'} D_{ii'}q_{i'}, \tag{17}$$

where $J_{ii'}$ and $D_{ii'}$ are real quantities independent of $\boldsymbol{q}$. When eqs. (15)–(17) are substituted into eq. (9), $\mathscr{E}$ is written as

$$\mathscr{E}=\Omega\sum_{ii'}\left\{\frac{1}{2}J_{ii'}(S_{1,i}S_{1,i'}+S_{2,i}S_{2,i'})+[\boldsymbol{S}_1\times\boldsymbol{S}_2]_i D_{ii'}q_{i'}\right\}. \tag{18}$$

The above expression for the spin-dependent energy was already used for the discussion of the stability of long-period helical spin structures in metals with no inversion center.[5,6)]

In the following sections, we derive $D_{\boldsymbol{G}\boldsymbol{G}',i}(\boldsymbol{q})$ and $D_{ii'}$ together with $J_{\boldsymbol{G}\boldsymbol{G}',ii'}(\boldsymbol{q})$ and $J_{ii'}$ from a microscopic Hamiltonian.

§3. Hamiltonian

We express the Hamiltonian of a metal in one-electron approximation in the following way:

$$\mathscr{H}=\mathscr{H}_0+\mathscr{H}_{\mathrm{so}}+\mathscr{H}_{\mathrm{ex}}+\mathscr{H}_{\mathrm{Z}}. \tag{19}$$

The first term $\mathscr{H}_0$ is the Hamiltonian without the spin-orbit coupling in the nonmagnetic phase:

$$\mathscr{H}_0=\frac{p^2}{2m}+V(\boldsymbol{r}), \tag{20}$$

where $p^2/2m$ is the kinetic energy of one electron and $V(\boldsymbol{r})$ is the one-electron potential taking account of the Coulomb and exchange interactions between electrons. The second term $\mathscr{H}_{\mathrm{so}}$ is the energy of the spin-orbit coupling:

$$\mathscr{H}_{\mathrm{so}}=\frac{\hbar}{4m^2c^2}[\nabla V(\boldsymbol{r})\times\boldsymbol{p}]\cdot\boldsymbol{\sigma}, \tag{21}$$

where σ_i denotes a Pauli matrix and other notations are usual. The third term $\mathscr{H}_{\mathrm{ex}}$ is the excess exchange energy which arises under the static spin polarization $\boldsymbol{S}(\boldsymbol{r})$. We employ the approximation that $\mathscr{H}_{\mathrm{ex}}$ is a local potential having the form as

$$\mathscr{H}_{\mathrm{ex}}=-V^{\mathrm{ex}}(\boldsymbol{r})\boldsymbol{S}(\boldsymbol{r})\cdot\boldsymbol{\sigma}. \tag{22}$$

Here, $V^{\mathrm{ex}}(\boldsymbol{r})$ is a periodic potential with the period of the lattice. One can know $V^{\mathrm{ex}}(\boldsymbol{r})$ by using the spin-polarized $X\alpha$ potential[17)] or the spin-polarized exchange correlation potential of the local spin density functional theory.[18)] The last term $\mathscr{H}_{\mathrm{Z}}$ is the Zeeman energy under a spatially varying magnetic field $\boldsymbol{H}(\boldsymbol{r})$:

$$\mathscr{H}_{\mathrm{Z}}=-\mu_{\mathrm{B}}\boldsymbol{H}(\boldsymbol{r})\cdot\boldsymbol{\sigma}, \tag{23}$$

where μ_{B} is the Bohr magneton.

We start with the eigenstates of $\mathscr{H}_0$. As was mentioned in §1, we are interested in a metal with no inversion center. For simplicity, we focus our attention on a single electronic band. The wave function $\psi_{\boldsymbol{k}}(\boldsymbol{r})$ of an electron with wave vector $\boldsymbol{k}$ and energy $\varepsilon_{\boldsymbol{k}}$ satisfies

$$\mathscr{H}_0\psi_{\boldsymbol{k}}(\boldsymbol{r})=\varepsilon_{\boldsymbol{k}}\psi_{\boldsymbol{k}}(\boldsymbol{r}), \tag{24}$$

$$\psi_{\boldsymbol{k}}(\boldsymbol{r})=\frac{1}{\sqrt{\Omega}}u_{\boldsymbol{k}}(\boldsymbol{r})\exp(\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}), \tag{25}$$

where $u_{\boldsymbol{k}}(\boldsymbol{r})$ is a periodic function with the period of the lattice. We introduce the annihilation operator $a_{\boldsymbol{k}s}$ and the creation operator $a^{\dagger}_{\boldsymbol{k}s}$ of the electron with wave vector $\boldsymbol{k}$ and spin $s\,(=\pm 1)$; the value 1 or -1 of s corresponds to the up or down spin respectively. In terms of $a_{\boldsymbol{k}s}$ and $a^{\dagger}_{\boldsymbol{k}s}$, the total Hamiltonian of the electron system $\mathscr{H}_{\mathrm{tot}}$ is expressed by

$$\begin{aligned}\mathscr{H}_{\mathrm{tot}}=&\sum_{\boldsymbol{k},s}\varepsilon_{\boldsymbol{k}}a^{\dagger}_{\boldsymbol{k}s}a_{\boldsymbol{k}s}+\sum_{\boldsymbol{k},ss'}\boldsymbol{L}_{\boldsymbol{k}}\cdot\boldsymbol{\sigma}_{ss'}a^{\dagger}_{\boldsymbol{k}s}a_{\boldsymbol{k}s'}-\sum_{\boldsymbol{q},\boldsymbol{G}}\sum_{\boldsymbol{k},ss'}\mu_{\mathrm{B}}F_{\boldsymbol{G}}(\boldsymbol{q},\boldsymbol{k})[\boldsymbol{H}_{\boldsymbol{G}}(\boldsymbol{q})+\boldsymbol{H}^{\mathrm{ex}}_{\boldsymbol{G}}(\boldsymbol{q})]\cdot\boldsymbol{\sigma}_{ss'}a^{\dagger}_{\boldsymbol{k}+\boldsymbol{q}s}a_{\boldsymbol{k}s'}\\&+\frac{1}{2}\sum_{\boldsymbol{q},\boldsymbol{G}}\sum_{\boldsymbol{k},ss'}\mu_{\mathrm{B}}F_{\boldsymbol{G}}(\boldsymbol{q},\boldsymbol{k})\boldsymbol{H}^{\mathrm{ex}}_{\boldsymbol{G}}(\boldsymbol{q})\cdot\boldsymbol{\sigma}_{ss'}\langle\mathrm{g}|a^{\dagger}_{\boldsymbol{k}+\boldsymbol{q}s}a_{\boldsymbol{k}s'}|\mathrm{g}\rangle,\end{aligned} \tag{26}$$

where

$$F_{\boldsymbol{G}}(\boldsymbol{q},\boldsymbol{k})=\sum_{\boldsymbol{G}'}u^{*}_{\boldsymbol{k}+\boldsymbol{q}+\boldsymbol{G}+\boldsymbol{G}'}u_{\boldsymbol{k}+\boldsymbol{G}'}, \tag{27}$$

and $\boldsymbol{L}_{\boldsymbol{k}}$ and $\boldsymbol{H}^{\mathrm{ex}}_{\boldsymbol{G}}(\boldsymbol{q})$ have the components as

$$L_{k,i}=\frac{\hbar}{4m^2c^2}\int \psi_k(r)^*[\nabla V(r)\times p]_i\psi_k(r)\,\mathrm{d}r, \tag{28}$$

$$H^{\mathrm{ex}}_{G,i}(q)=\frac{1}{\mu_{\mathrm{B}}}\sum_{G'} S_{G-G',i}(q)V^{\mathrm{ex}}_{G'}. \tag{29}$$

The Fourier components $H_{G,i}(q)$, $V^{\mathrm{ex}}_G(q)$ and u_{k+G} in eqs. (26)–(29) have been defined by

$$H_i(r)=\sum_{q,G} H_{G,i}(q)\exp\{\mathrm{i}(q+G)\cdot r\}, \tag{30}$$

$$V^{\mathrm{ex}}(r)=\sum_G V^{\mathrm{ex}}_G\exp(\mathrm{i}G\cdot r), \tag{31}$$

$$u_k(r)=\sum_G u_{k+G}\exp(\mathrm{i}G\cdot r). \tag{32}$$

The calculations are made at zero temperature in the following. The last term in eq. (26) is $(-1/2)$ times the expected value of the exchange energy $\mathscr{H}_{\mathrm{ex}}$ in the ground state $|\mathrm{g}\rangle$, which should be added for avoiding its double counting. As shown in Appendix I, L_k with a general k does not vanish in a metal with no inversion center. The first two terms on the right hand side of eq. (26) are diagonalized when we introduce c_{ks} given by

$$\begin{bmatrix} c_{k1} \\ c_{k-1}\end{bmatrix}=\mathbf{U}_k\begin{bmatrix} a_{k1} \\ a_{k-1}\end{bmatrix}, \tag{33}$$

$$\mathbf{U}_k=\frac{1}{\sqrt{2L_k(L_k-L_{k,z})}}\begin{bmatrix} L_{k,x}+\mathrm{i}L_{k,y}, & L_k-L_{k,z} \\ L_k-L_{k,z}, & -L_{k,x}+\mathrm{i}L_{k,y}\end{bmatrix}, \tag{34}$$

with

$$L_k=\sqrt{L^2_{k,x}+L^2_{k,y}+L^2_{k,z}}. \tag{35}$$

Then, the Hamiltonian (26) is transformed into

$$\mathscr{H}_{\mathrm{tot}}=\sum_{k,s}(\varepsilon_k+sL_k)c^\dagger_{ks}c_{ks}-\sum_{q,G}\sum_{k,ss'}\mu_{\mathrm{B}}F_G(q,k)H^{\mathrm{eff}}_G(q)\cdot(\mathbf{U}_{k+q}\sigma\mathbf{U}_k)_{ss'}c^\dagger_{k+qs}c_{ks'}$$
$$+\frac{1}{2}\sum_{q,G}\sum_{k,ss'}\mu_{\mathrm{B}}F_G(q,k)H^{\mathrm{ex}}_G(q)\cdot(\mathbf{U}_{k+q}\sigma\mathbf{U}_k)_{ss'}\langle \mathrm{g}|c^\dagger_{k+qs}c_{ks'}|\mathrm{g}\rangle, \tag{36}$$

where $H^{\mathrm{eff}}_G(q)$ is the effective magnetic field given by

$$H^{\mathrm{eff}}_G(q)=H_G(q)+H^{\mathrm{ex}}_G(q). \tag{37}$$

As can be seen from eq. (36), the new bands with $s=1$ and $s=-1$ correspond, respectively, to the upper and lower bands among the two bands split by the spin-orbit coupling.

§4. Wave-Vector-Dependent Susceptibility Affected by the Spin-Orbit Coupling

We derive here the wave-vector-dependent susceptibility which plays an important role in the calculation of the bilinear term of the spin density. The expected value of the spin density in the ground electronic state,

$$S_G(q)=\frac{1}{2\Omega}\sum_{k,ss'}F_G(q,k)^*(\mathbf{U}_k\sigma\mathbf{U}_{k+q})_{s's}\langle \mathrm{g}|c^\dagger_{ks'}c_{k+qs}|\mathrm{g}\rangle, \tag{38}$$

becomes nonvanishing under the effective magnetic fields $H^{\mathrm{eff}}_G(q)$. Regarding the second term on the right hand side of eq. (36) as a perturbation on the first term, we can obtain the wave function of the ground state to its first order. After calculating the expected value in eq. (38) by use of the wave functions thus obtained, we have

$$2\mu_{\mathrm{B}}S_{G,i}(q)=\sum_{G',i'}\chi^0_{GG',ii'}(q)^*H^{\mathrm{eff}}_{G',i'}(q). \tag{39}$$

Here, the susceptibility unenhanced by the exchange interaction, $\chi^0_{GG',ii'}(q)$, is expressed as follows:

$$\chi^{0}_{GG',ii'}(q)=\chi^{0I}_{GG'}(q)\delta_{ii'}+\frac{1}{2}\sum_{k,ss'}\lambda_{GG',ss'}(q,k)(\mathbf{U}_{k+q}\sigma_i\mathbf{U}_k^{-1})_{ss'}(\mathbf{U}_k\sigma_{i'}\mathbf{U}_{k+q})_{s's},\tag{40}$$

with

$$\chi^{0I}_{GG'}(q)=\frac{2}{\Omega}\sum_{k,ss'}\mu_B^2F_G(q,k)F_{G'}(q,k)^*\left[\frac{\theta(\varepsilon_F-\varepsilon_k)-\theta(\varepsilon_F-\varepsilon_{k+q})}{\varepsilon_{k+q}-\varepsilon_k}\right],\tag{41}$$

and

$$\lambda_{GG',ss'}(q,k)=\frac{2}{\Omega}\mu_B^2F_G(q,k)F_{G'}(q,k)^*\left[\frac{\theta(\varepsilon_F-\varepsilon_k-s'L_k)-\theta(\varepsilon_F-\varepsilon_{k+q}-sL_{k+q})}{(\varepsilon_{k+q}+sL_{k+q})-(\varepsilon_k+s'L_k)}-\frac{\theta(\varepsilon_F-\varepsilon_k)-\theta(\varepsilon_F-\varepsilon_{k+q})}{\varepsilon_{k+q}-\varepsilon_k}\right],\tag{42}$$

where $\theta(x)$ is 1 for $x>0$ and 0 for $x<0$, ε_F denotes the Fermi level, and $\delta_{ii'}$ is the Kronecker δ function. The first term on the right hand side of eq. (40) represents the isotropic susceptibility in the absence of the spin-orbit coupling and the second term represents a change in the susceptibility caused by the presence of the spin-orbit coupling. The susceptibilities enhanced by the exchange interaction, $\chi_{GG',ii'}(q)$, are defined by

$$2\mu_B S_{G,i}(q)=\sum_{G',i'}\chi_{GG',ii'}(q)^*H_{G',i'}(q).\tag{43}$$

Equations (37), (39) and (43) reveal that the susceptibilities $\chi_{GG',ii,}(q)$ satisfy

$$\sum_{G'',i''}\chi_{GG'',ii''}(q)\left[\delta_{G''G'}\delta_{i''i'}-\frac{1}{2\mu_B^2}\sum_{G'''}V^{ex*}_{G''-G'''}\chi^{0}_{G'''G',i''i'}(q)\right]=\chi^{0}_{GG',ii'}(q).\tag{44}$$

Next, we go into detail of the expression of $\chi^0_{GG',ii'}(q)$ given by eq. (40). It can be easily proved that

$$\lambda_{GG',ss'}(q,k)=\lambda^{(1)}_{GG'}(q,k)+\lambda^{(2)}_{GG'}(q,k)\sigma_{z,ss}\sigma_{z,s's'}+\lambda^{(3)}_{GG'}(q,k)\sigma_{z,ss}+\lambda^{(4)}_{GG'}(q,k)\sigma_{z,s's'},\tag{45}$$

where

$$\begin{bmatrix}\lambda^{(1)}_{GG'}(q,k)\\ \lambda^{(2)}_{GG'}(q,k)\\ \lambda^{(3)}_{GG'}(q,k)\\ \lambda^{(4)}_{GG'}(q,k)\end{bmatrix}=\frac{1}{4}\begin{bmatrix}1&1&1&1\\1&-1&-1&1\\1&1&-1&-1\\1&-1&1&-1\end{bmatrix}\begin{bmatrix}\lambda_{GG',11}(q,k)\\ \lambda_{GG',1-1}(q,k)\\ \lambda_{GG',-11}(q,k)\\ \lambda_{GG',-1-1}(q,k)\end{bmatrix}.\tag{46}$$

Substituting eq. (45) into eq. (40) and using the relation

$$\mathbf{U}_k^{-1}\sigma_z\mathbf{U}_k=L_k\cdot\sigma/L_k\ ,\tag{47}$$

we find that $\chi^0_{GG',ii'}(q)$ consists of three terms:

$$\chi^{0}_{GG',ii'}(q)=\chi^{0I}_{GG'}(q)\delta_{ii'}+\chi^{0S}_{GG',ii'}(q)+\chi^{0A}_{GG',ii'}(q),\tag{48}$$

where $\chi^{0S}_{GG',ii'}(q)$ and $\chi^{0A}_{GG',ii'}(q)$ are respectively symmetric and antisymmetric with respect to the interchange of i and i'. The latter two terms are given by

$$\chi^{0S}_{GG',ii'}(q)(=\chi^{0S}_{GG',i'i}(q))=\sum_k\left\{\lambda^{(1)}_{GG'}(q,k)\delta_{ii'}+\lambda^{(2)}_{GG'}(q,k)\left[\frac{L_{k+q,i}L_{k,i'}+L_{k+q,i'}L_{k,i}}{L_{k+q}L_k}-\frac{L_{k+q}\cdot L_k}{L_{k+q}L_k}\delta_{ii'}\right]\right\},\tag{49}$$

$$\chi^{0A}_{GG',ii'}(q)(=-\chi^{0A}_{GG',i'i}(q))=\mathrm{i}\sum_k\left\{\lambda^{(3)}_{GG'}(q,k)\frac{L_{k+q,i''}}{L_{k+q}}-\lambda^{(4)}_{GG'}(q,k)\frac{L_{k,i''}}{L_k}\right\}.\tag{50}$$

The susceptibilities $\chi^{0S}_{GG',ii'}(q)$ and $\chi^{0A}_{GG',ii'}(q)$ begin respectively with the second order and the first order of the spin-orbit couplings L_k in their expansions in powers of the couplings L_k. Further, we notice that $\chi^{0S}_{GG',ii'}(q)=\chi^{0S}_{G'G,ii'}(-q)$ and $\chi^{0A}_{GG',ii'}(q)=-\chi^{0A}_{G'G,ii'}(-q)$.

When $\boldsymbol{G}=\boldsymbol{G}'=0$ and q is sufficiently small compared with reciprocal lattice vectors, the expressions for the susceptibilities (41), (49) and (50) can be more simplified. In general, the spin-orbit splitting is much smaller than ε_F so that $(\varepsilon_F-\varepsilon_{\boldsymbol{k}}-\xi_{\boldsymbol{k}})$ with $\xi_{\boldsymbol{k}}=\varepsilon_{\boldsymbol{k}+\boldsymbol{q}}-\varepsilon_{\boldsymbol{k}}+sL_{\boldsymbol{k}+\boldsymbol{q}}$ or $\xi_{\boldsymbol{k}}=s'L_{\boldsymbol{k}}$ on the right hand side of eq. (42) is expanded in powers of $\xi_{\boldsymbol{k}}$. By using a formula for $\theta(\varepsilon_F-\varepsilon_{\boldsymbol{k}}-\xi_{\boldsymbol{k}})$ which is given in Appendix II and also the normalization condition for $\psi_{\boldsymbol{k}}(\boldsymbol{r})$, $F_0(0,\boldsymbol{k})=1$, $\lambda_{00,ss'}(\boldsymbol{q},\boldsymbol{k})$ is shown to be given by

$$\lambda_{00,ss'}(\boldsymbol{q},\boldsymbol{k})$$
$$=\frac{\mu_B^2}{\Omega}\left\{-sL_{\boldsymbol{k}+\boldsymbol{q}}-s'L_{\boldsymbol{k}}+\frac{1}{3}\left[L_{\boldsymbol{k}+\boldsymbol{q}}^2+L_{\boldsymbol{k}}^2+ss'L_{\boldsymbol{k}+\boldsymbol{q}}L_{\boldsymbol{k}}+(2sL_{\boldsymbol{k}+\boldsymbol{q}}+s'L_{\boldsymbol{k}})(\varepsilon_{\boldsymbol{k}+\boldsymbol{q}}-\varepsilon_{\boldsymbol{k}})\frac{\partial}{\partial\varepsilon_F}\right]\right\}\frac{\partial^2}{\partial\varepsilon_F^2}\theta(\varepsilon_F-\varepsilon_{\boldsymbol{k}}). \tag{51}$$

We substitute eq. (51) into eqs. (49) and (50), and expand further the susceptibilities in powers of $\boldsymbol{q}$ up to its first order to have

$$\chi_{00}^{0\mathrm{I}}(\boldsymbol{q})=2\mu_B^2\mathscr{D}(\varepsilon_F), \tag{52}$$

$$\chi_{00,ii'}^{0\mathrm{S}}(\boldsymbol{q})=\frac{2\mu_B^2}{3\Omega}\frac{\partial^3}{\partial\varepsilon_F^3}\sum_{\boldsymbol{k}}\left(L_{\boldsymbol{k},i}L_{\boldsymbol{k},i'}+\frac{1}{2}L_{\boldsymbol{k}}^2\delta_{ii'}\right)\theta(\varepsilon_F-\varepsilon_{\boldsymbol{k}}), \tag{53}$$

$$\chi_{00,ii'}^{0\mathrm{A}}(\boldsymbol{q})=\frac{\mathrm{i}\mu_B^2}{\Omega}\left\{-\frac{\partial^2}{\partial\varepsilon_F^2}\sum_{\boldsymbol{k}}\left(\frac{\partial L_{\boldsymbol{k},i''}}{\partial\boldsymbol{k}}\cdot\boldsymbol{q}\right)\theta(\varepsilon_F-\varepsilon_{\boldsymbol{k}})+\frac{1}{3}\frac{\partial^3}{\partial\varepsilon_F^3}\sum_{\boldsymbol{k}}L_{\boldsymbol{k},i''}\left(\frac{\partial\varepsilon_{\boldsymbol{k}}}{\partial\boldsymbol{k}}\cdot\boldsymbol{q}\right)\theta(\varepsilon_F-\varepsilon_{\boldsymbol{k}})\right\}, \tag{54}$$

where $\mathscr{D}(\varepsilon_F)$ is the density of states at ε_F per unit volume for the up or down spin.

As will be seen in the next section, the appearance of the antisymmetric susceptibility is vital to the ASI.

§5. Spin-Dependent Energy in the Metal with the Spin-Orbit Coupling

The aim of this section is to find the energy of the system bilinear in the spin-polarization densities $\boldsymbol{S}_{\boldsymbol{G}}(\boldsymbol{q})$. In order to construct the spin polarizations, we introduce a fictitious, spatially varying field $\boldsymbol{H}(\boldsymbol{r})$ or its Fourier transforms $\boldsymbol{H}_{\boldsymbol{G}}(\boldsymbol{q})$ whose effect together with the effect of the exchange field $\boldsymbol{H}_{\boldsymbol{G}}^{\mathrm{ex}}(\boldsymbol{q})$ on the wave functions can be calculated up to their first orders by the perturbation theory. The change of the total energy due to the spin polarizations, $\mathscr{E}$, is then obtained by calculating the expected value of $\mathscr{H}_{\mathrm{tot}}$ by use of the wave functions. Note that the Zeeman energy of the fictitious field is not included in $\mathscr{H}_{\mathrm{tot}}$. The resulting $\mathscr{E}$ has the following form:

$$\mathscr{E}=\frac{\Omega}{2}\sum_{\boldsymbol{q}}\sum_{\boldsymbol{G}\boldsymbol{G}'}\sum_{ii'}H_{\boldsymbol{G},i}^{\mathrm{eff}}(\boldsymbol{q})\chi_{\boldsymbol{G}\boldsymbol{G}',ii'}(\boldsymbol{q})H_{\boldsymbol{G}',i'}(\boldsymbol{q})^*. \tag{55}$$

Since the magnetic fields $H_{\boldsymbol{G},i}^{\mathrm{eff}}(\boldsymbol{q})$ and $H_{\boldsymbol{G},i}(\boldsymbol{q})$ are related to the spin densities $S_{\boldsymbol{G},i}(\boldsymbol{q})$ through eqs. (39) and (43), eq. (55) is rewritten as

$$\mathscr{E}=2\mu_B^2\Omega\sum_{\boldsymbol{q}}\sum_{\boldsymbol{G}\boldsymbol{G}'}\sum_{ii'}S_{\boldsymbol{G},i}(\boldsymbol{q})[\chi(\boldsymbol{q})]_{\boldsymbol{G}\boldsymbol{G}',ii'}^{-1}S_{\boldsymbol{G}',i'}(\boldsymbol{q})^*, \tag{56}$$

where $[\chi(\boldsymbol{q})]^{-1}$ represents the inverse of the matrix $[\chi(\boldsymbol{q})]$ with the elements $\chi_{\boldsymbol{G}\boldsymbol{G}',ii'}(\boldsymbol{q})$, i.e.,

$$\sum_{\boldsymbol{G}'',i''}[\chi(\boldsymbol{q})]_{\boldsymbol{G}\boldsymbol{G}'',ii''}^{-1}\chi_{\boldsymbol{G}''\boldsymbol{G}',i''i'}(\boldsymbol{q})=\delta_{\boldsymbol{G}\boldsymbol{G}'}\delta_{ii'}. \tag{57}$$

Equations (44) and (56) together with eq. (57) lead us to

$$\mathscr{E}=2\mu_B^2\Omega\sum_{\boldsymbol{q}}\sum_{\boldsymbol{G}\boldsymbol{G}'}\sum_{ii'}S_{\boldsymbol{G},i}(\boldsymbol{q})[\chi^0(\boldsymbol{q})]_{\boldsymbol{G}\boldsymbol{G}',ii'}^{-1}S_{\boldsymbol{G}',i'}(\boldsymbol{q})^*-\Omega\sum_{\boldsymbol{q}}\sum_{\boldsymbol{G}\boldsymbol{G}'}V_{\boldsymbol{G}-\boldsymbol{G}'}^{\mathrm{ex}*}[\boldsymbol{S}_{\boldsymbol{G}}(\boldsymbol{q})\cdot\boldsymbol{S}_{\boldsymbol{G}'}(\boldsymbol{q})^*], \tag{58}$$

with the matrix $[\chi^0(\boldsymbol{q})]$ possessing the elements $\chi_{\boldsymbol{G}\boldsymbol{G}',ii'}^0(\boldsymbol{q})$.

The inverse susceptibility $[\chi^0(\boldsymbol{q})]_{\boldsymbol{G}\boldsymbol{G}',ii'}^{-1}$ can be divided into three terms:

$$[\chi^0(\boldsymbol{q})]_{\boldsymbol{G}\boldsymbol{G}',ii'}^{-1}=[\chi^{0\mathrm{I}}(\boldsymbol{q})]_{\boldsymbol{G}\boldsymbol{G}'}^{-1}\delta_{ii'}+[\chi^0(\boldsymbol{q})]_{\boldsymbol{G}\boldsymbol{G}',ii'}^{-1\mathrm{S}}+[\chi^0(\boldsymbol{q})]_{\boldsymbol{G}\boldsymbol{G}',ii'}^{-1\mathrm{A}}, \tag{59}$$

where $[\chi^{0\mathrm{I}}(\boldsymbol{q})]$ is the matrix with the elements $\chi_{\boldsymbol{G}\boldsymbol{G}'}^{0\mathrm{I}}(\boldsymbol{q})$, and $[\chi^0(\boldsymbol{q})]_{\boldsymbol{G}\boldsymbol{G}',ii'}^{-1\mathrm{S}}$ and $[\chi^0(\boldsymbol{q})]_{\boldsymbol{G}\boldsymbol{G}',ii'}^{-1\mathrm{A}}$ are respectively symmetric and antisymmetric with respect to the interchange of i and i'. The latter two terms in eq. (59) are given to the first order of the susceptibilities $\chi^{0S}_{GG',ii'}(\boldsymbol{q})$ and $\chi^{0A}_{GG',ii'}(\boldsymbol{q})$ defined by eqs. (49) and (50) by

$$[\chi^0(\boldsymbol{q})]^{-1S}_{GG',ii'}(=[\chi^0(\boldsymbol{q})]^{-1S}_{GG',i'i})=-\sum_{G''G'''}[\chi^{0I}(\boldsymbol{q})]^{-1}_{GG''}\chi^{0S}_{G''G''',ii'}(\boldsymbol{q})[\chi^{0I}(\boldsymbol{q})]^{-1}_{G'''G'}, \tag{60}$$

$$[\chi^0(\boldsymbol{q})]^{-1A}_{GG',ii'}(=-[\chi^0(\boldsymbol{q})]^{-1A}_{GG',i'i})=-\sum_{G''G'''}[\chi^{0I}(\boldsymbol{q})]^{-1}_{GG''}\chi^{0A}_{G''G''',ii'}(\boldsymbol{q})[\chi^{0I}(\boldsymbol{q})]^{-1}_{G'''G'}. \tag{61}$$

When eq. (59) is substituted into eq. (58), we obtain $\mathscr{E}$ which has the same form as eq. (9). A comparison between the two gives

$$J_{GG',ii'}(\boldsymbol{q})=\{2\mu_B^2[\chi^{0I}(\boldsymbol{q})]^{-1}_{GG'}-V^{ex}_{G-G'}\}\delta_{ii'}-2\mu_B^2\sum_{G''G'''}[\chi^{0I}(\boldsymbol{q})]^{-1}_{GG''}\chi^{0S}_{G''G''',ii'}(\boldsymbol{q})[\chi^{0I}(\boldsymbol{q})]^{-1}_{G'''G'}, \tag{62}$$

$$D_{GG',i}(\boldsymbol{q})=-2\mu_B^2\sum_{G''G'''}[\chi^{0I}(\boldsymbol{q})]^{-1}_{GG''}\chi^{0A}_{G''G''',i'i''}(\boldsymbol{q})[\chi^{0I}(\boldsymbol{q})]^{-1}_{G'''G'}. \tag{63}$$

The first term on the right hand side of eq. (62) represents the spin interaction constant in the absence of the spin-orbit coupling. If this term is negative, a spin polarization appears spontaneously. The fourth order terms of the spin densities in the spin-dependent energy will arrest the developments of the spin polarizations. The second term in eq. (62) is due to the presence of the spin-orbit coupling, and begins with the second order terms of the spin-orbit coupling constants $\boldsymbol{L_k}$ if expanded. In a crystal (or spin) structure of low symmetry, the second term causes the magnetic anisotropy. The ASI constant given by eq. (63) is also due to the presence of the spin-orbit coupling, and begins with the first order terms of the spin-orbit coupling constants. It should be noticed that the exchange interactions $V^{ex}_{G-G'}(\boldsymbol{q})$ does not play any role in the ASI obtained here.

We are again interested in $J_{00,ii'}(\boldsymbol{q})$ and $D_{00,i}(\boldsymbol{q})$ with a small q. When the terms with the non-vanishing reciplocal lattice vectors in eqs. (62) and (63) are all neglected, the expressions for $J_{00,ii'}(\boldsymbol{q})$ and $D_{00,i'}(\boldsymbol{q})$ can be simplified by use of eqs. (52)–(54) as follows:

$$J_{00,ii'}(\boldsymbol{q})(\simeq J_{ii'})=[\mathscr{D}(\varepsilon_F)^{-1}-V^{ex}_0]\delta_{ii'}-\frac{1}{3\Omega\mathscr{D}(\varepsilon_F)^2}\frac{\partial^3}{\partial\varepsilon_F^3}\sum_{\boldsymbol{k}}\left(L_{\boldsymbol{k},i}L_{\boldsymbol{k},i'}+\frac{1}{2}L_{\boldsymbol{k}}^2\delta_{ii'}\right)\theta(\varepsilon_F-\varepsilon_{\boldsymbol{k}}), \tag{64}$$

$$D_{00,i}(\boldsymbol{q})=\frac{i}{3\Omega\mathscr{D}(\varepsilon_F)^2}\frac{\partial^2}{\partial\varepsilon_F^2}\sum_{\boldsymbol{k}}\left(\frac{\partial L_{\boldsymbol{k},i}}{\partial\boldsymbol{k}}\cdot\boldsymbol{q}\right)\theta(\varepsilon_F-\varepsilon_{\boldsymbol{k}}). \tag{65}$$

In obtaining eq. (65), we have used a partial integration. Equations (17) and (65) give

$$D_{ii'}=-\frac{1}{3\Omega\mathscr{D}(\varepsilon_F)^2}\frac{\partial^2}{\partial\varepsilon_F^2}\sum_{\boldsymbol{k}}\left(\frac{\partial L_{\boldsymbol{k},i}}{\partial k_{i'}}\right)\theta(\varepsilon_F-\varepsilon_{\boldsymbol{k}}). \tag{66}$$

The sum on the right hand side of eqs. (65) and (66) can be confined to those $\boldsymbol{k}$ on the Fermi surface. As is shown in Appendix I, the integrand is an even function of $\boldsymbol{k}$ whose average value over the Fermi surface does not vanish in general.

In the following we discuss an order-of-magnitude estimate of the antisymmetric interaction in the case of MnSi. The ionic positions in MnSi are given by (u, u, u), $(\frac{1}{2}+u, \frac{1}{2}-u, -u)$, $(-u, \frac{1}{2}+u, \frac{1}{2}-u)$ and $(\frac{1}{2}-u, -u, \frac{1}{2}+u)$ where u is 0.138 for Mn ions and 0.845 for Si ions. Considering an overall cubic symmetry of the crystal, we conclude $D_{ii'}=D\delta_{ii'}$ which leads us to the expression $D[\boldsymbol{S}_1\times\boldsymbol{S}_2]\cdot\boldsymbol{q}$ for the second term on the right hand side of eq. (18). We adapt eq. (66) to the present case to take into account the fact that several bands are partially filled. Since the interband matrix elements of the spin-orbit coupling turns out to be small compared with the intraband ones, we neglect the formers to obtain the expression as

$$D=-\frac{1}{3\Omega\mathscr{D}(\varepsilon_F)^2}\frac{\partial^2}{\partial\varepsilon_F^2}\sum_{\boldsymbol{k},\nu}\left(\frac{\partial L_{\boldsymbol{k}\nu,i}}{\partial k_i}\right)\theta(\varepsilon_F-\varepsilon_{\boldsymbol{k}\nu}), \tag{67}$$

where ν is the band index. According to the self-consistent APW band calculation of MnSi which was already reported,[19] five bands are involved in the calculation of D. In order to obtain a rough order-of-magnitude estimate, we simplify the calculation to replace the integrand for each band in eq. (67) by its value at the $\boldsymbol{k}$ in the $\langle 111 \rangle$ direction, where the vectors $\boldsymbol{L}_{\boldsymbol{k}\nu}$ are all parallel to the direction of $\boldsymbol{k}$. We obtain then the numerical value of D given by

$$D = -0.97 \text{ meV}(a/2\pi),$$

with the lattice constant $a = 4.548$ Å.[20] This value of D should be compared with $|D| = 0.775 \text{ meV}(a/2\pi)$ deduced from the observed value of $q = 0.025(2\pi/a)$ of the helical structure.[4] The apparent success of our numerical estimate, however, should be checked by a more detailed calculation in future. Nevertheless, we may conclude at least that our discussion on the underlying mechanism of the antisymmetric interaction is quantitatively meaningful in MnSi, i.e., a representative compound exhibiting itinerant electron magnetism.

§6. Concluding Remarks

In the above, we have started with a Bloch band in the nonmagnetic phase which is split by the spin-orbit coupling because of the lack of the inversion symmetry, and have studied the effects of spin polarizations on the energy of the band electron system. It has been shown that the resulting energy change of the system due to the spin polarizations contains the ASI owing to the existence of the spin-orbit coupling. In magnetic insulators, the ASI constant is roughly $\Delta g/g$ ($\ll 1$) times the SSI constant, where g is the gyromagnetic ratio and Δg is its deviation from the value of a free electron.[16) This is because the ASI in magnetic insulators originates from an energy change due to the mixing between two localized levels split by the crystal field through both the spin-orbit coupling and the superexchange (or direct exchange) interaction. In a metal with no inversion center, however, the expected value of $\boldsymbol{L} = \hbar[\nabla V(\boldsymbol{r}) \times \boldsymbol{p}]/4m^2c^2$ in a Bloch state $\psi_{\boldsymbol{k}}$, $\boldsymbol{L}_{\boldsymbol{k}}$, does not vanish. Therefore, the energy of the ASI is nothing else but a part of the energy of the spin-orbit coupling in the presence of the spin polarizations. This implies that in the metal under consideration, any reduction factor such as $\Delta g/g$ in insulators does not appear in the ratio of the ASI constant to the SSI constant, and the ASI energy is sometimes allowed to exceed the SSI energy. We expect, however, that the ASI energy is considerably smaller than the spin-orbit coupling energy in ionic states itself because only the potential parts with no inversion symmetry can contribute to the ASI with the aid of the spin-orbit coupling.

In this paper, we have investigated the intraband effect of the spin-orbit coupling on the spin-dependent energy. Even under this simplification, we have been able to see some aspects of the ASI in metals. In real metals, of course, there exist multi bands and the mixings between different bands through the spin-orbit coupling give additional terms to the expression of the ASI. The extension of our calculation to the case of a more general metal is left for a future work.

Appendix I

We here summarize the properties of $\boldsymbol{L}_{\boldsymbol{k}}$ which is defined by eq. (28). Since $\boldsymbol{L}$ is an hermite operator, $\boldsymbol{L}_{\boldsymbol{k}}$ is real. When there exists no band degeneracy, $\psi_{\boldsymbol{k}}(\boldsymbol{r})^*$ equals $\psi_{-\boldsymbol{k}+\boldsymbol{G}}(\boldsymbol{r})$ where $\boldsymbol{G}$ is an arbitrary reciprocal lattice vector including $\boldsymbol{G}=0$. This gives

$$\boldsymbol{L}_{\boldsymbol{k}} = -\boldsymbol{L}_{-\boldsymbol{k}+\boldsymbol{G}}, \tag{A·1}$$

together with $\varepsilon_{\boldsymbol{k}} = \varepsilon_{-\boldsymbol{k}}$. If a metal has an inversion center, $\psi_{\boldsymbol{k}}(\boldsymbol{r})$ equals $\psi_{-\boldsymbol{k}+\boldsymbol{G}}(-\boldsymbol{r})$ and $\boldsymbol{L}$ is invariant by the inversion. Therefore, we have

$$\boldsymbol{L}_{\boldsymbol{k}} = \boldsymbol{L}_{-\boldsymbol{k}+\boldsymbol{G}}. \tag{A·2}$$

Equations (A·1) and (A·2) prove that $\boldsymbol{L}_{\boldsymbol{k}}$ with any $\boldsymbol{k}$ vanishes in a metal with an inversion center. In a metal with no inversion center, $\boldsymbol{L}_{\boldsymbol{k}}$ vanishes only at $\boldsymbol{k} = \boldsymbol{G}/2$, and $(\partial \boldsymbol{L}_{\boldsymbol{k},i}/\partial \boldsymbol{k}_{i'})$ can have a nonvanishing value at any $\boldsymbol{k}$. It is noted that both $(\partial L_{\boldsymbol{k},i}/\partial k_{i'})$ and $(L_{\boldsymbol{k},i}\partial\varepsilon_{\boldsymbol{k}}/\partial k_{i'})$ are even functions of $\boldsymbol{k}$.

Appendix II

In obtaining eq. (51), we have used the formula for $\theta(\varepsilon_{\rm F} - \varepsilon_{\boldsymbol{k}} - \xi_{\boldsymbol{k}})$ as follows:

$$\theta(\varepsilon_{\rm F} - \varepsilon_{\boldsymbol{k}} - \xi_{\boldsymbol{k}}) \simeq \sum_{n=0}^{n_{\rm f}} \frac{(-\xi_{\boldsymbol{k}})^n}{n!} \frac{\partial^n \theta(\varepsilon_{\rm F} - \varepsilon_{\boldsymbol{k}})}{\partial \varepsilon_{\rm F}^n}, \tag{A·3}$$

where $\xi_{\boldsymbol{k}}$ has small values compared with $\varepsilon_{\rm F}$ and $n_{\rm f}$ is a sufficiently large integer. For simplicity, we give a proof of eq. (A·3) in the simplest case of the one dimension and $n_{\rm f} = 1$.

The proof given below can be easily extended to a general case. Let us consider a quantity $\mathscr{G}$ given by

$$\mathscr{G}=\int_{-\infty}^{\infty} g\,(k)\theta(\varepsilon_{\mathrm{F}}-\varepsilon_k-\xi_k)\,\mathrm{d}k$$
$$=\int_{-\infty}^{k_{\mathrm{F}}} g(k)\,\mathrm{d}k. \qquad (\mathrm{A}\cdot4)$$

Here, $g(k)$ is an arbitrary function of k and k_{F} is assumed to be only one solution of

$$\varepsilon_{\mathrm{F}}-\varepsilon_{k_{\mathrm{F}}}-\xi_{k_{\mathrm{F}}}=0. \qquad (\mathrm{A}\cdot5)$$

When k_{F}^0 satisfies $\varepsilon_{\mathrm{F}}=\varepsilon_{k_{\mathrm{F}}^0}$, k_{F} is approximated as

$$k_{\mathrm{F}}=k_{\mathrm{F}}^0-\left(\frac{\partial\varepsilon_{k_{\mathrm{F}}^0}}{\partial k_{\mathrm{F}}^0}\right)\xi_{k_{\mathrm{F}}^0}. \qquad (\mathrm{A}\cdot6)$$

From eqs. (A·4) and (A·6), we obtain

$$\mathscr{G}=\int_{-\infty}^{k_{\mathrm{F}}^0} g(k)\,\mathrm{d}k-g(k_{\mathrm{F}}^0)\xi_{k_{\mathrm{F}}^0}\left(\frac{\partial\varepsilon_{k_{\mathrm{F}}^0}}{\partial k_{\mathrm{F}}^0}\right)^{-1}. \qquad (\mathrm{A}\cdot7)$$

This equation is changed to be

$$\mathscr{G}=\int_{-\infty}^{\infty} g(k)\theta(\varepsilon_{\mathrm{F}}-\varepsilon_k)\,\mathrm{d}k$$
$$-\int_{-\infty}^{\infty} g(k)\xi_k\delta(\varepsilon_{\mathrm{F}}-\varepsilon_k)\left(\frac{\partial\varepsilon_k}{\partial k}\right)^{-1}\mathrm{d}\varepsilon_k$$
$$=\int_{-\infty}^{\infty} g(k)\left[1-\xi_k\frac{\partial}{\partial\varepsilon_{\mathrm{F}}}\right]\theta(\varepsilon_{\mathrm{F}}-\varepsilon_k)\,\mathrm{d}k, \qquad (\mathrm{A}\cdot8)$$

where $\delta(x)$ is the Dirac δ function. Equations (A·4) and (A·8) prove eq. (A·3) of this case.

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